

Mixed Decimal Review — Estimate It, Then Nail It

Answer Key

Grade 5

Quick Answers

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|--|--|
| 1. 30 | 8. 4.25 |
| 2. 5 | 9. 6 |
| 3. 6.15 | 10. 3.6 |
| 4. 5.65 | 11. (a) comparison symbol = $>$, — |
| 5. 0.21 | 12. (a) estimate = 8, (b) exact cost = 8.19, (c) |
| 6. 12 | change = 1.81 |
| 7. (a) estimate = 12, (b) exact cost = 12.60 | |
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What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 11. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 5

5.NBT.B.7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

3. *If they got 3.99: lined up the right-hand digits instead of the decimal points (added 24 hundredths instead of 2.4)*

5. *If they got 2.1: counted only one decimal place in the product*

6. *If they got 1.2: shifted the decimal point in only one of the two numbers*

1. Estimate: 6.1×4.8

Solution: Round each factor to the nearest whole number: $6.1 \rightarrow 6$ and $4.8 \rightarrow 5$. Then $6 \times 5 = 30$.

30

2. Estimate: $19.6 \div 4.1$

Solution: Round each number: $19.6 \rightarrow 20$ and $4.1 \rightarrow 4$. Then $20 \div 4 = 5$.

5

3. Add: $3.75 + 2.4$

Solution: Line up the *decimal points*, not the last digits: 2.4 is 2.40. Then $3.75 + 2.40 = 6.15$.

6.15

4. Subtract: $8 - 2.35$

Solution: Write 8 as 8.00, then subtract with regrouping: $8.00 - 2.35 = 5.65$.

5.65

5. Multiply: 0.7×0.3

Solution: $7 \times 3 = 21$; the factors carry two decimal places between them, so the product is 21 hundredths.

0.21

6. Divide: $9.6 \div 0.8$

Solution: Shift *both* numbers one place so the divisor is whole: $96 \div 8 = 12$.

12

7. Markers: 4 packs at 3.15 dollars

Solution: (a) 3.15 rounds to 3, and 3×4 gives the estimate.

12

(b) Exact: $3.15 \times 4 = 12 + 0.60$, so the total is dollars — close to the estimate, so it is reasonable.

12.60

8. Sandwich 2.35, juice 3.4, paid with 10 dollars

Solution: First the total: $2.35 + 3.40 = 5.75$. Then the change: $10.00 - 5.75 = 4.25$.

4.25

9. Plank: 5.4 m cut into 0.9-m shelves

Solution: “How many 0.9s fit in 5.4?” Shift both one place: $54 \div 9 = 6$. So the plank makes shelves.

6

10. Strawberries: 2.4 dollars per kg, 1.5 kg

Solution: 2.4×1.5 : as whole numbers $24 \times 15 = 360$; two decimal places between the factors gives 3.60.

3.6

11. Ella’s estimate: 3.2×4.1 vs. 12

Solution: (a) The exact product is greater:

$$3.2 \times 4.1 > 12$$

(b) Ella rounded *both* factors down (3.2 down to 3, 4.1 down to 4), so her estimate multiplied two smaller numbers. The true product must therefore be *greater* than 12. A correct explanation names the downward rounding of both factors; no exact computation is needed.

12. Challenge: apples 2.1 dollars/kg, 3.9 kg, 10-dollar bill

Solution: (a) Round: $3.9 \rightarrow 4$ and $2.1 \rightarrow 2$, so the cost is about 4×2 dollars.

$$8$$

(b) Exact: 3.9×2.1 : as whole numbers $39 \times 21 = 819$; two decimal places gives 8.19 dollars.

$$8.19$$

(c) Change: $10.00 - 8.19 = 1.81$ dollars — and $10 - 8 = 2$ from the estimate confirms it is reasonable.

$$1.81$$